

8 (a) (i) potential divider method gives C1

$$V = 9.0 \left(\frac{470}{470 + 1650} \right)$$

$$V = 2.00 \text{ V} \quad \text{A1}$$

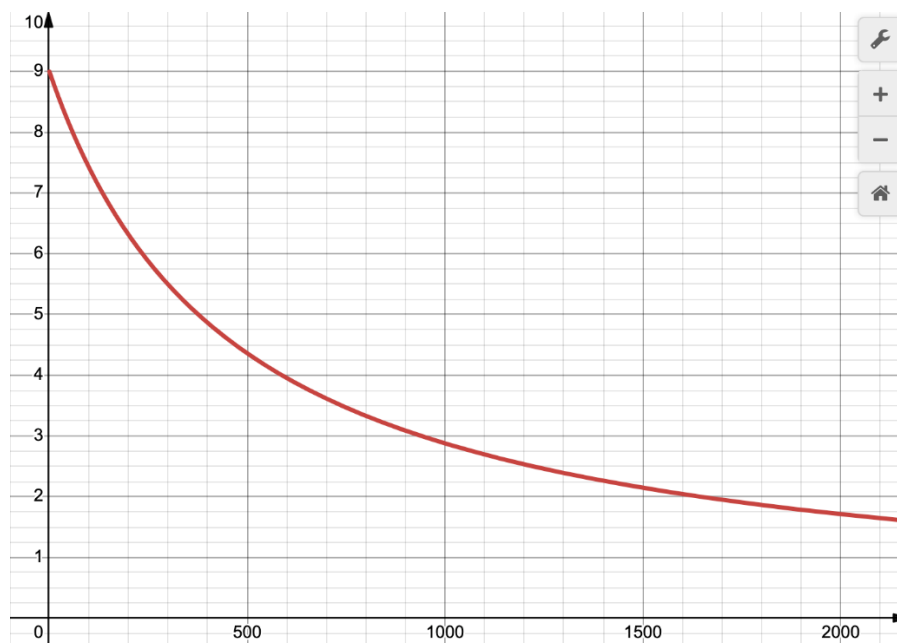
(ii) downward sloping with decreasing gradient B1

cuts at (0,9) B1

and graph below 2 V (or the value in (a)(i)) when curve reaches $R_1 = 2000 \Omega$

Example

voltmeter reading / V



Justification

$$V = 9 \left(\frac{470}{470 + R} \right)$$

R_1 / Ω

(b) (i) $\frac{1}{R_{\parallel}} = \frac{1}{470} + \frac{1}{100\,000}$, $R_{\parallel} = 467.8 \Omega$ B1

$$V = 9 \left(\frac{467.8}{467.8 + 1650} \right) = 1.988 \quad \text{B1}$$

$$\frac{1.988-1.955}{1.955} \times 100\% = -0.35\% \quad \text{A1}$$

- (ii) No significant changes / very slight differences / voltmeter may not have enough precision to detect the difference. B1
 comparison using numbers. B1

E.g.

internal resistance is 3 orders of magnitude smaller than resistance in circuit.

Or, take internal resistance to be maximum at $10 \, \Omega$, fraction

$$\approx \frac{10}{1645+640} = 0.004.$$

internal resistance is small compared to resistance in circuit (1 mark maximum)